

CSE 401: Numerical Analysis, Simulation and Modeling

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Introduction, Approximations & Errors

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Last Compiled: June 21, 2026

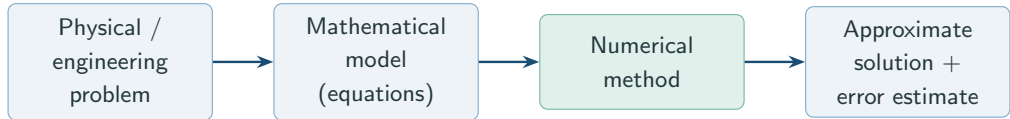
*Everything you compute over the next few weeks will be wrong.
This course spends an uncomfortable amount of time asking how wrong, and
whether wrong is still useful.*

The big idea

Engineering questions become *mathematical models*. A lot of useful models cannot be solved with pencil-and-paper formulas, so we solve them *approximately* — that approximation is numerical methods.

Calculus promises exact answers; numerical methods are what you do when calculus stops returning your calls.

From a physical problem to a number

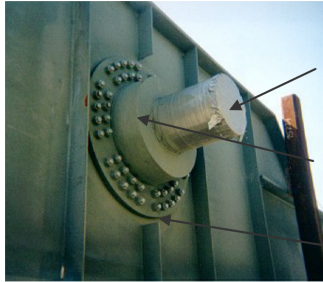


Bascule-Bridge

A common type of movable drawbridge that opens to allow boat traffic to pass underneath



The bascule-bridge trunnion



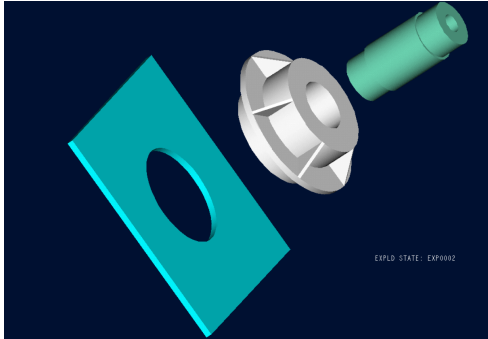
Trunnion

Hub

Girder

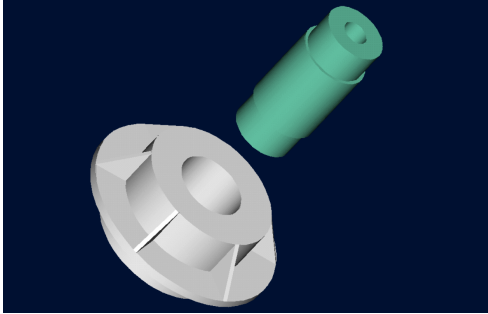
A hollow steel *trunnion* is shrink-fit into a *hub*, then into the bridge *girder*.

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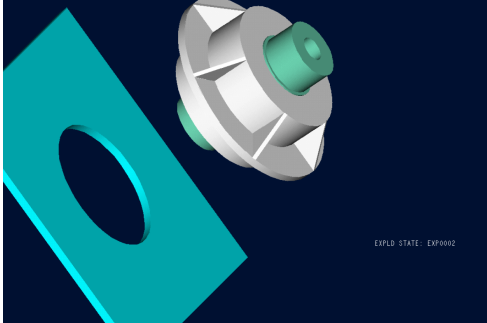
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A hollow steel *trunnion* is shrink-fit into a *hub*, then into the bridge *girder*.

- Chill the trunnion $\Rightarrow$ it contracts $\Rightarrow$ slide it through the hub.

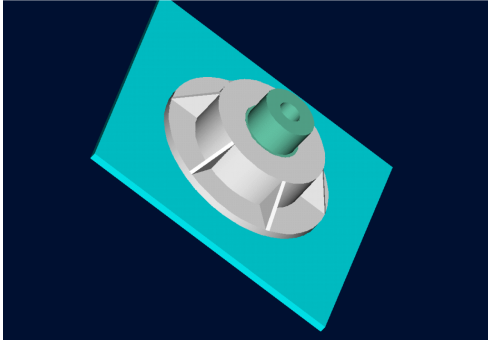
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- On warming it expands $\Rightarrow$ a tight fit.

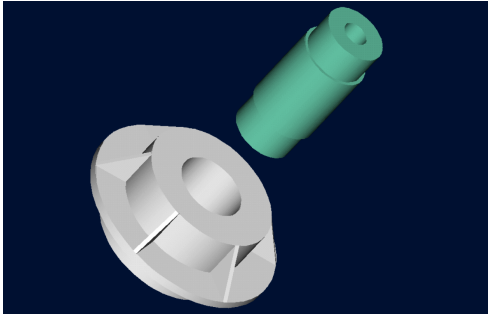
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- Trunnion-hub chilled and then pressed into the girder, where they warm-up to a tight fit.

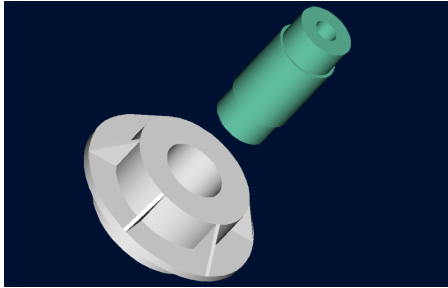
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- Chill the trunnion $\Rightarrow$ it contracts $\Rightarrow$ slide it through the hub.
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- Trunnion-hub chilled and then pressed into the girder, where they warm-up to a tight fit.

Florida, 1995: the trunnion *got stuck* mid-insertion. A replacement trunnion + hub would cost \$50,000; with delays, a six-figure loss. *The trunnion was pulled out before it stuck permanently.*



Trunnion OD = 12.363'', hub ID = 12.358'', required diametric clearance = 0.01''. Dry-ice/alcohol bath at -108°F ; room temperature 80°F .

Required contraction

$$12.363 - 12.358 + 0.01 = 0.015''$$

Question. Will immersion in dry-ice/alcohol contract the trunnion by at least 0.015''?

First model — constant expansion coefficient

Using a single (room-temperature) coefficient α :

$$\Delta D = D \alpha \Delta T, \quad \Delta T = T_{\text{fluid}} - T_{\text{room}} = -108 - 80 = -188^\circ\text{F}.$$

Example (plug in the numbers). With $D = 12.363''$ and $\alpha = 6.47 \times 10^{-6} \frac{\text{in}}{\text{in}^\circ\text{F}}$:

$$\Delta D = (12.363)(6.47 \times 10^{-6})(-188) = -0.01504''.$$

Looks fine on paper

Predicted contraction $0.01504'' > 0.015''$ required — so the bath *should* work... yet the trunnion got stuck. What went wrong: a bad model, or a calculation error?

But α varies with T

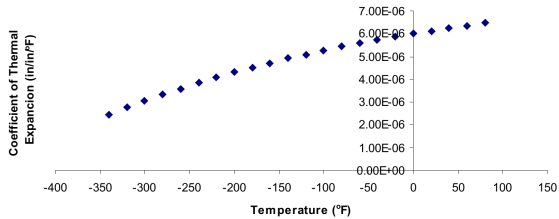


Table 1 (abridged).

T (°F)	α ($\mu\text{in}/\text{in}/^\circ\text{F}$)
80	6.47
40	6.24
0	6.00
-40	5.72
-100	5.28
-160	4.72
-220	4.08
-300	3.07
-340	2.45

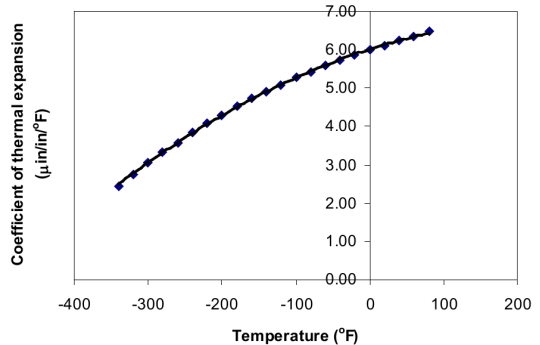
Because $\alpha = \alpha(T)$, the contraction is an *integral*, not a product:

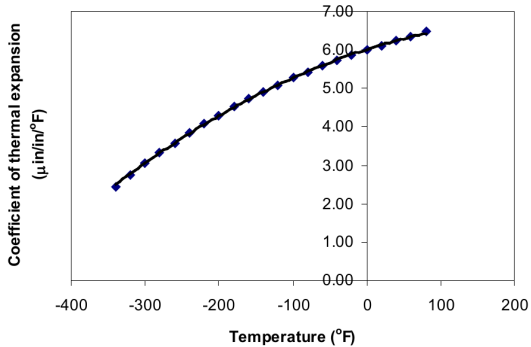
$$\Delta D = D \int_{T_{\text{room}}}^{T_{\text{fluid}}} \alpha(T) dT.$$

How to get $\alpha(T)$? Curve fitting

We assume a second-order polynomial.

$$\alpha(T) = a_0 + a_1 T + a_2 T^2, \quad \begin{aligned} a_0 &= 6.0150 \times 10^{-6} \\ a_1 &= 6.1946 \times 10^{-9} \\ a_2 &= -1.2278 \times 10^{-11} \end{aligned}$$





Carrying out the integral with the fitted $\alpha(T)$:

$$\Delta D = -0.013689''.$$

Getting stuck was predictable

Actual contraction $0.013689'' < 0.015''$ (required).
The trunnion didn't shrink enough, so got stuck.
The constant- α formula was wrong.

We need good **models** and **tools** to solve them, and we need to know **how wrong** the answers are, so we can decide if they are good enough.

Numerical answers are *approximate*, so error is unavoidable. To handle it we:

1. **identify** where error comes from,
2. **quantify** it, and
3. **minimise** it to our needs.

Step (2): quantify

We need numbers that say *how wrong* an approximation is and, crucially, numbers we can compute *even when the true answer is unknown*.

Definition

$$E_t = \text{True value} - \text{Approximate value.}$$

Example 1. Approximate $f'(2)$ for $f(x) = 7e^{0.5x}$ by a forward difference with $h = 0.3$:

$$f'(x) \approx \frac{f(x+h) - f(x)}{h} \Rightarrow f'(2) \approx \frac{7e^{0.5(2.3)} - 7e^{0.5(2)}}{0.3} = \frac{22.107 - 19.028}{0.3} = 10.265.$$

Exact value: $f'(x) = 3.5e^{0.5x} \Rightarrow f'(2) = 3.5e^1 = 9.5140$. Hence

$$E_t = 9.5140 - 10.265 = -0.75061.$$

The minus sign: we estimated too high.

Magnitude alone is misleading

An error of $E_t = -0.75061$ *sounds* small.

Same problem, rescaled. If instead $f(x) = 7 \times 10^{-6} e^{0.5x}$, every value shrinks by 10^{-6} , so

$$E_t = -0.75061 \times 10^{-6}.$$

The problem is *identical* (same x , same h), yet the raw error looks 10^6 times “better”.

We need an error *relative* to the size of the answer.

Definition

$$\epsilon_t = \frac{\text{True error}}{\text{True value}} = \frac{E_t}{\text{True value}}.$$

Example 2. For Example 1,

$$\epsilon_t = \frac{-0.75061}{9.5140} = -0.078895 = -7.8895\%.$$

As an *absolute relative true error*: $|\epsilon_t| = 0.078895 = 7.8895\%$.

Approximate error E_a — when the truth is unknown

We rarely know the true value (if we did, why approximate?).

We try to find solutions *iteratively* until they converge (*good enough*).

In an *iterative* or *refining* calculation we compare successive estimates.

Definition

$$E_a = \text{Present approximation} - \text{Previous approximation.}$$

Example 3. Same derivative, two step sizes:

$$h = 0.3 : f'(2) \approx 10.265, \quad h = 0.15 : f'(2) \approx \frac{7e^{0.5(2.15)} - 7e^1}{0.15} = 9.8799.$$

$$E_a = 9.8799 - 10.265 = -0.38474.$$

Relative approximate error ϵ_a

Definition

$$\epsilon_a = \frac{\text{Approximate error}}{\text{Present approximation}} = \frac{E_a}{\text{Present approximation}}.$$

Example 4. From Example 3,

$$\epsilon_a = \frac{-0.38474}{9.8799} = -0.038942 = -3.8942\%, \quad |\epsilon_a| = 3.8942\%.$$

ϵ_a is computable *without the true value*, this is the quantity we actually monitor.

Two ways to stop an iteration

(i) Pre-specified tolerance ϵ_s

Stop when $|\epsilon_a| \leq \epsilon_s$. Choose ϵ_s to match the accuracy you need.

(ii) Pre-specified significant digits m

To trust at least m significant digits in the answer, iterate until

$$|\epsilon_a| \leq 0.5 \times 10^{2-m} \% .$$

Example 5

The Maclaurin series: For a function $f(x)$ with derivatives at $x = 0$:

$$f(x) = f(0) + f'(0)x + \frac{f''(0)}{2!}x^2 + \frac{f'''(0)}{3!}x^3 + \cdots = \sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!}x^n.$$

Estimate $e^{0.7}$ from the Maclaurin series $e^x = 1 + x + \frac{x^2}{2!} + \cdots$

6 terms: $e^{0.7} \approx 2.0136$, 5 terms: $e^{0.7} \approx 2.0122$.

$$|\epsilon_a| = \left| \frac{2.0136 - 2.0122}{2.0136} \right| \times 100\% = 0.069527\%.$$

Test against $m = 2$: $0.5 \times 10^{2-2}\% = 0.5 \times 10^0\% = 0.5\%$.

Conclusion

Since $0.069527\% \leq 0.5\%$, **at least 2 significant digits are correct** if we use 6 terms

What is a significant digit?

Significant digits report *how much you trust* a number.

Example. “The county population is **1 million**”, only *one* significant digit (1).

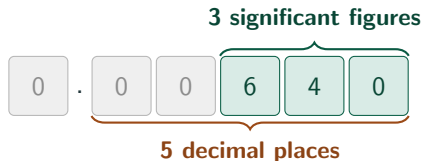
The exact count 1,079,587 has *seven*.

Scientific notation makes the difference explicit:

$$1,000,000 = 1 \times 10^6 \quad (1 \text{ s.f.}), \quad 1,079,587 = 1.079587 \times 10^6 \quad (7 \text{ s.f.}).$$

Decimal places $\neq$ significant figures

- **Decimal places** — count every digit *after* the point.
- **Significant figures** — meaningful digits, *starting at the first non-zero one* (let's not worry about leading zeros).



Normalised scientific notation: $|x| = m \times 10^n$

Every non-zero number can be written

$$|x| = m \times 10^n, \quad 1 \leq m < 10, \quad n = \lfloor \log_{10} |x| \rfloor.$$

Forcing $1 \leq m < 10$ puts *one* non-zero digit before the point $\Rightarrow$ 1st sig. digit at 10^n place

1st	2nd	3rd	4th	5th
3	7	2	5	6
10^3	10^2	10^1	10^0	10^{-1}

$3725.6 = 3.7256 \times 10^3$, so $m = 3.7256$, $n = 3$; the leading 3 sits in the 10^3 (thousands) place

Where the M -th significant digit lives

$$\underbrace{10^n}_{1\text{st}}, \underbrace{10^{n-1}}_{2\text{nd}}, \underbrace{10^{n-2}}_{3\text{rd}}, \dots, \boxed{\underbrace{10^{n-M+1}}_{M\text{-th}}}.$$

Core fact

The M -th significant digit sits in the 10^{n-M+1} place.

Example. For $3725.6 = 3.7256 \times 10^3$ ($n = 3$), the 3rd significant digit (the 2) sits in the $10^{3-3+1} = 10^1$ place, the tens column.

The claim

If $\left| \frac{x - \hat{x}}{x} \right| \leq 0.5 \times 10^{2-M} \%$, then $\hat{x}$ is correct to at least M significant digits.

Since “%” means “ $\times 10^{-2}$ ”, the same bound as a plain fraction is

$$0.5 \times 10^{2-M} \% = 0.5 \times 10^{2-M} \times 10^{-2} = \frac{1}{2} \times 10^{-M}.$$

Example. For $M = 3$, the threshold: $0.5 \times 10^{-1}\% = 0.05\%$ (i.e. $\frac{1}{2} \times 10^{-3} = 0.0005$)

Step 1: “Correct to M digits” meaning

Definition

$\hat{x}$ is correct to M significant digits when the error is **at most** *half a unit in the place of the M -th significant digit*:

$$|x - \hat{x}| \leq \frac{1}{2} \times 10^{n-M+1}.$$

This is exactly the error you make when you correctly *round* x (the true value) to M figures

Example. For $\pi = 3.14159 \dots$ we have $m = 3.14159$, $n = 0$.

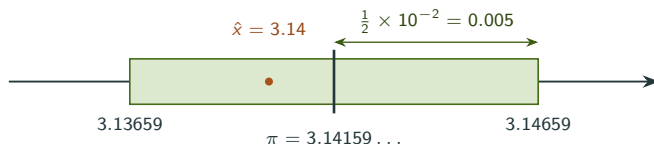
Correct to $M = 3$ figures means $|x - \hat{x}| \leq \frac{1}{2} \times 10^{0-3+1} = \frac{1}{2} \times 10^{-2} = 0.005$.

The rounding $\hat{x} = 3.14$ has error $|3.14159 - 3.14| = 0.00159 \leq 0.005$ (a unit in the third ($10^{0-3+1} = 10^{-2}$) place: 0.01, half of that: 0.005)

so 3.14 is correct to 3 significant figures

Step 1: “Correct to M figures” meaning

“Correct to M figures” is an *absolute* window of half-width $\frac{1}{2} \times 10^{n-M+1}$ around x . For π with $M = 3$ that half-width is 0.005:



Any $\hat{x}$ inside the green band agrees with π to 3 significant figures; 3.14 sits comfortably inside.

Step 2: Equivalence of the two definitions

$$\text{Let } \left| \frac{x - \hat{x}}{x} \right| \leq \frac{1}{2} \times 10^{-M}$$

$$\iff |x - \hat{x}| = \left| \frac{x - \hat{x}}{x} \right| \cdot m 10^n \leq \frac{1}{2} \times 10^{-M} \cdot m 10^n \text{ (Multiply by } |x| = m 10^n \text{)}$$

$$\iff |x - \hat{x}| = \left| \frac{x - \hat{x}}{x} \right| \cdot m 10^n \leq \frac{m}{2} \times 10^{n-M}$$

$$\iff |x - \hat{x}| < \frac{10}{2} \times 10^{n-M} = 5 \times 10^{n-M} = \frac{1}{2} \times 10^{n-M+1}, (m < 10)$$

$\frac{1}{2} \times 10^{n-M+1}$ is exactly the window in Step 1, so $\hat{x}$ is correct to at least M significant digits. ■

Where do errors come from?

- **Modelling error.** Wrong assumptions.
Example: assuming a quantity $\propto v$ when it is $\propto v^2$.
- **Data / measurement error** and **programming mistakes.**

Inherent to numerical methods themselves

Even with a perfect model and a perfect program, two errors remain:

(1) round-off error (2) truncation error.

Round-off error

Definition

A computer stores numbers with *finite* precision, so most numbers are kept only approximately.

Example. On a machine that keeps 6 digits, $\frac{1}{3}$ is stored as 0.333333:

$$\text{round-off error} = \frac{1}{3} - 0.333333 = 0.000000\bar{3} = 3.3333 \dots \times 10^{-7}.$$

Numbers like π and $\sqrt{2}$ are irrational

Never have an exact finite representation; approximated in every calculation

When I say - “I’ll be there in five minutes.”

Truncation error

Definition

The error from *truncating* (cutting short) a mathematical procedure.

Example. The Maclaurin series for $e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!}$ has infinitely many terms; keeping three gives

$$e^x \approx 1 + x + \frac{x^2}{2!}, \quad \text{truncation error} = e^x - \left(1 + x + \frac{x^2}{2!}\right) = \sum_{n=3}^{\infty} \frac{x^n}{n!}$$

The discarded “tail” is the truncation error.

Controlling truncation with ϵ_a

Computing $|\epsilon_a|$ for $e^{1.2} = \sum_{k \geq 0} \frac{1.2^k}{k!}$ and stop once # significant digits ≥ 2 .
 $|\epsilon_a| \leq 0.5 \times 10^{2-m}\%$ $\Rightarrow$ at least m significant digits.

n	S_n	E_a	$ \epsilon_a \%$	#s.d.
1	1.0000	—	—	—
2	2.2000	1.2	54.546	0
3	2.9200	0.72	24.658	0
4	3.2080	0.288	8.9776	0
5	3.2944	0.0864	2.6226	1
6	3.3151	0.020736	0.62550	1
7	3.3193	0.0041472	0.12494	2

stop: $0.12494\% \leq 0.5\%$ at $n = 7$.

Step 7 — add $\frac{1.2^6}{6!} = 0.0041472$.

$$S_7 = 3.3151 + 0.0041472 = 3.3193,$$

$$E_a = 0.0041472,$$

$$|\epsilon_a| = \left| \frac{0.0041472}{3.3193} \right| \times 100\% = 0.12494\%.$$

$0.12494\% \leq 0.5\%$ ($= 0.5 \times 10^0\%$) $\Rightarrow m = 2$. Bar cleared —
stop with at least **2** significant digits.

Truncation isn't only “chopping a series”

To find the derivative, we define: $f'(x) = \lim_{\Delta x \rightarrow 0} \frac{f(x+\Delta x) - f(x)}{\Delta x}$

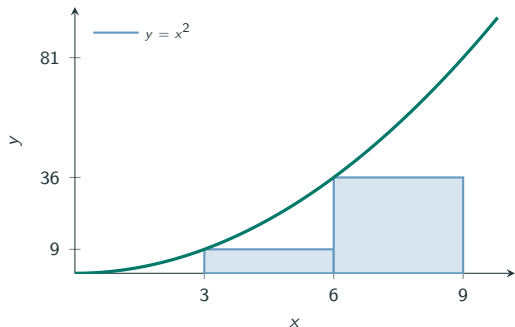
But when finite Δx used: $f'(x) \approx \frac{f(x+\Delta x) - f(x)}{\Delta x}$

Example. For $f(x) = x^2$, the exact $f'(3) = 2(3) = 6$

With $\Delta x = 0.2$:

$$f'(3) \approx \frac{(3.2)^2 - (3)^2}{0.2} = \frac{10.24 - 9}{0.2} = 6.2, \quad \text{truncation error} = 6 - 6.2 = -0.2.$$

Truncation in numerical integration



Exact area:

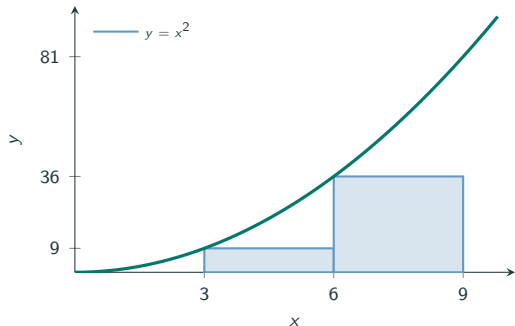
$$\int_3^9 x^2 dx = \left[\frac{x^3}{3} \right]_3^9 = \frac{729 - 27}{3} = 234.$$

Two left-endpoint rectangles (heights 9 and 36, width 3):

$$9 \cdot 3 + 36 \cdot 3 = 135.$$

$$\text{truncation error} = 234 - 135 = 99 (99/234 \approx 0.423).$$

Truncation in numerical integration



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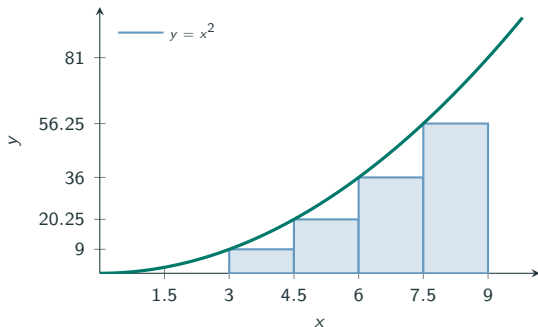
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truncation error = $234 - 135 = 99$ ($99/234 \approx 0.423$).

The white gap between curve and rectangles is the error.

Truncation in numerical integration



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Two left-endpoint rectangles (heights 9 and 36, width 3):

$$9 \cdot 3 + 36 \cdot 3 = 135.$$

truncation error = $234 - 135 = 99$ ($99/234 \approx 0.423$).

The white gap between curve and rectangles is the error.

Four left-endpoint rectangles (width 1.5):

$$1.5 \times (9 + 20.25 + 36 + 56.25) = 182.25.$$

trunc. error = $234 - 182.25 = 51.75$ ($51.75/234 \approx 0.221$).

More rectangles $\Rightarrow$ smaller gap $\Rightarrow$ less error.